

MULTI-AGENT DATA ANALYSIS AND INSIGHT GENERATION PLATFORM FOR FINANCIAL AND SALES DATA

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Presentation Outline

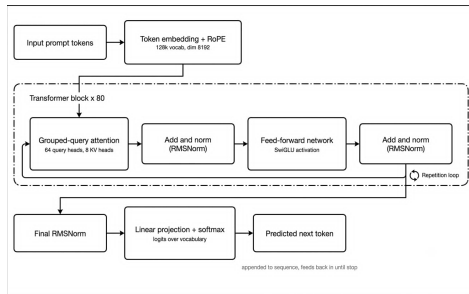
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Introduction

- The project is being implemented as a state-based data analysis pipeline for sales and financial CSV datasets.
- Four pipeline stages are currently connected: structural profiling, semantic tagging, preprocessing, and statistical analysis.
- The system transforms raw records into a cleaned dataset, a preprocessing audit trail, quality metrics, statistics, and chart files.
- Mathematical operations are performed with Python libraries; the language model is used for semantic interpretation and processing rules.
- The current implementation provides a working backend foundation for the planned user-facing platform.

LLM Architecture – Llama Transformer

- The model used in the pipeline is Llama, a decoder-only transformer language model.
- It processes the input context through repeated self-attention and feed-forward blocks.
- The attention layers help the model relate each token to earlier tokens in the prompt.
- This architecture makes it suitable for semantic tagging, schema inference, and structured reasoning.



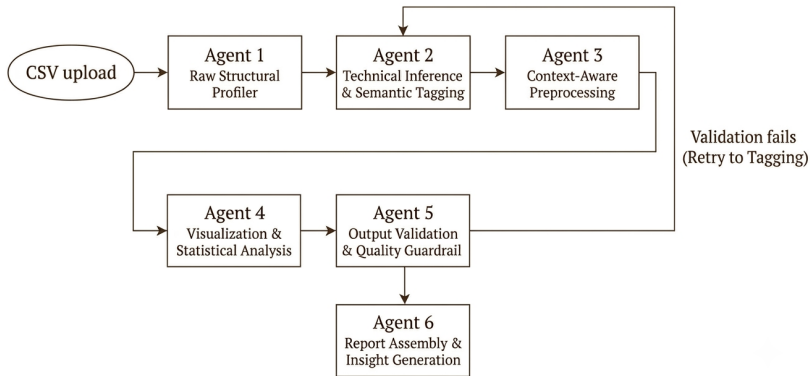
Objectives

- Develop an autonomous multi-agent system that converts raw sales and financial data into reproducible analytical outputs.
- Implement semantic, context-aware preprocessing instead of applying the same cleaning rule to every column.
- Evaluate the pipeline using representative datasets, focused unit tests, and generated diagnostics.
- Establish a reliable backend foundation before completing report generation and the user interface.

Scope of Project

- Current implementation: sales and financial datasets in CSV format
- Completed: structural profiling, semantic tagging, preprocessing, and pipeline routing
- Completed: data quality scoring, audit logging, derived business metrics, and chart generation
- In progress: richer statistical analysis
- Future scope: web interface and support for Excel, JSON, and Parquet inputs

Methodology – Current 4-Stage Pipeline



Agent 1 – Raw Structural Profiler

- Reads the input CSV and records shape, dtypes, missingness, uniqueness, and duplicates.
- Produces a raw structural profile without changing the source data.
- Caches the DataFrame for the downstream stages.

[Agent 1] Profiled: 10000 rows x 21 cols | Missing: 0.0% | Duplicates: 0

Agent 2 – Technical Inference & Semantic Tagging

- Sniffs numeric, datetime, boolean, and string types using local heuristics.
- Uses Groq-based semantic tagging when available and falls back to deterministic rules when it is not.
- Creates a per-column blueprint containing type, semantic tag, scaling, imputation, and analysis policies.
- Passes the blueprint to preprocessing instead of hard-coding one rule for every column.

```
country
semantic_tag      : geographic
intended_type     : string
imputation        : mode
encoding          : one_hot
identifier        : False
analysis_allowed  : True
confidence        : 60.0
null_action       : impute_mode
suitable          : True
reason            : low_cardinality_category
```

Agent 3 – Context-Aware Preprocessing

- Coerces columns according to the semantic blueprint and cleans international currency formats.
- Standardizes text and null variants while preserving identifiers.
- Removes duplicates, applies configured imputation, and clips permitted outliers with IQR bounds.
- Extracts date features and derives metrics such as profit, margin, and revenue per unit.
- Records scaling parameters, an audit log, and a 0–100 data quality score.

```
[Agent 3] Preprocessing: profile-strict domain-finance sales input-10000x21  
[Agent 3] completed: rows-10000->10000 cols-21->115 quality-100.0/100 row_missing=0.0% remaining_nulls=0.0%  
[Agent 3] cleaning summary: dodged=0, scaled=1, added cols=94, removed rows=0  
[Agent 3] Note: the cleaned dataset is normalized and exported to outputs/cleaned_data.csv; if the row count stays the same  
as to drop.  
[Agent 4] Starting analysis: 10000 rows x 115 cols
```

Agent 4 – Visualization & Statistical Analysis

- Calculates descriptive statistics, correlations, distributions, growth, and seasonality.
- Detects anomalies using a configurable z-score threshold.
- Computes regression trends and category rankings when the dataset supports them.
- Saves reproducible PNG charts and structured statistics to the pipeline state.

```
[Agent 4] Starting analysis: 10000 rows x 115 cols  
[Agent 4] Step 1 – Descriptive stats: 27 columns  
[Agent 4] Step 2 – Correlation done, strong pairs: 8  
[Agent 4] Step 3 – Growth rates skipped (no revenue/time axis)  
[Agent 4] Step 4 – Rankings skipped (no revenue/category pairing)  
[Agent 4] Step 5 – Seasonality skipped (no time axis)  
[Agent 4] Step 6 – Anomalies: 235 unique rows (2.35%) across 1 column  
[Agent 4] Step 7 – Distributions done (4 charts)  
[Agent 4] Step 8 – Regression done (27 columns, 0 charts)  
[Agent 4] Step 9 – Distribution charts done (1 charts)  
[Agent 4] Step 10 – Derived metrics skipped (no derived metrics)  
[Agent 4] Done – 6 charts saved to outputs/charts/
```

Mid-Progress Timeline

- **Completed:** project architecture, GraphState, Agents 1–3, conditional pipeline routing, preprocessing audit trail, quality scoring, and focused tests.
- **Current:** Agent 4 statistical analysis and chart generation, verified on the Amazon sales dataset.
- **Next:** connect statistics to evidence-backed insight generation.
- **Final phase:** upload workflow, frontend presentation, broader datasets, and end-to-end evaluation.

Remaining Work

Work package	Target outcome
Insight generation	Produce traceable natural-language findings
Interface	Upload data and display progress/results
Evaluation	Benchmark accuracy, robustness, and run-time

Conclusion

- **Backend foundation achieved:** A working LangGraph pipeline now carries a CSV dataset through four specialized stages.
- **Core transformation achieved:** Profiling, semantic tagging, context-aware preprocessing, quality scoring, and statistical chart generation are operational.
- **Evidence available:** The verified run produces cleaned data, diagnostics, audit records, statistics, and chart artifacts.
- **Next objective:** complete validation, evidence-backed report generation, interface integration, and broader evaluation.

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